

The elongation is computed as:

$$\text{Elongation} = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}}$$

- If the tumor is roughly spherical, $\lambda_{\max} \approx \lambda_{\min}$, so elongation ≈ 1 .
- If the tumor is stretched along one axis, $\lambda_{\max} \gg \lambda_{\min}$, so elongation becomes large.

$$\text{Sphericity} = \frac{\pi^{1/3} (6V)^{2/3}}{A}$$

- Sphericity ≈ 1 for a perfect sphere.
- Values decrease below 1 as the tumor becomes less spherical (e.g., irregular or elongated shapes).

$$\text{Compactness} = \frac{V^2}{A^3}$$

- Compactness is highest for sphere-like tumors.
- Values decrease toward 0 for shapes with large surface area relative to their volume (thin, elongated, or highly irregular tumor).
- V : tumor volume
- A : surface area of the tumor

$$\text{Aspect Ratio} = \frac{\text{AP}}{\text{SI}}$$

- If ≈ 1 , the tumor is about as tall (SI) as it is deep (AP).
- If Aspect Ratio > 1.1 , the tumor is elongated front-back compared to top-bottom.

$$d_{\text{midline}} = |c_x - c_{0x}|$$

- c_x : x-coordinate of the centroid (world coordinates).
- c_{0x} : x-coordinate of the image center (world coordinates).

$$d_{\text{AP}} = \|\mathbf{c} - \mathbf{c}_0\| \cos(\theta_{\text{AP}}) \cdot s_{\text{AP}}$$

- d_{AP} : signed displacement along the anteroposterior (AP) axis.
- $\mathbf{c}$: tumor centroid in world coordinates.
- $\mathbf{c}_0$: image center in world coordinates.
- θ_{AP} : angle between displacement vector $(\mathbf{c} - \mathbf{c}_0)$ and the AP axis unit vector $\mathbf{u}_{\text{AP}}$.
- s_{AP} : sign factor (+1 posterior, -1 anterior).

Similarly, d_{SI} : signed displacement along the superior-inferior (SI) axis.